

Statistical Simulation in ngspice

Monte Carlo, Latin-Hypercube, high-sigma, correlations, yield, model-declared statistics, and the record of every draw

Ngspice + OpenVAF Enhancements project

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Statistical simulation in ngspice — a complete guide

This build of ngspice-46 has a full statistical-analysis suite: ordinary Monte Carlo, Latin-Hypercube low-discrepancy sampling, high-sigma rare-event estimation, native process/mismatch correlations, a packaged yield command with per-sample recording and tracking, worst-case distance, design centring — and, since Enhancement-530, statistics that live in the Verilog-A model itself: a parameter declares its own spread with an attribute and the simulator draws it, trial after trial, with no `gauss()` in the deck. Every draw of every run can be written to a file beside the netlist, with the results computed from that run on the same row.

This note documents all of it end-to-end — the distribution functions, the sampling controls, the commands, the model-side attributes, the recording — with worked examples whose outputs are taken from real runs of the committed binary, so you can go from "I have a Verilog-A/SPICE circuit" to "here is its yield, its 5-sigma failure probability, the draws behind every sample, and a file with all of it."

Two sources of variation exist, and they compose:

where the statistics live	how they are drawn	who re-draws
the netlist: <code>agauss()</code> , <code>gauss()</code> , <code>unif()</code> , <code>aunif()</code> , <code>limit()</code> , <code>mvnorm()</code> in a <code>.param</code> , a device value, a subcircuit call	ngspice's netlist PRNG, once per <i>sample</i>	a <code>reset</code> , or the loop commands' fast path (§2)

where the statistics live	how they are drawn	who re-draws
the model: (* std=... *) and friends on a Verilog-A parameter, under .option osdimc	a pure hash of (mcseed, trial, owner, parameter) — no RNG state	every run-class command is a <i>trial</i> (§7)

Everything on the netlist side is a front-end capability: it changes only which values the random .params take, not the circuit solve, so it works identically under the Sparse 1.3 and KLU linear solvers and with built-in *and* OSDI/Verilog-A devices. The model side writes each draw through the ordinary parameter setter — the alter path — so it, too, leaves the solve untouched.

TL;DR — what's available

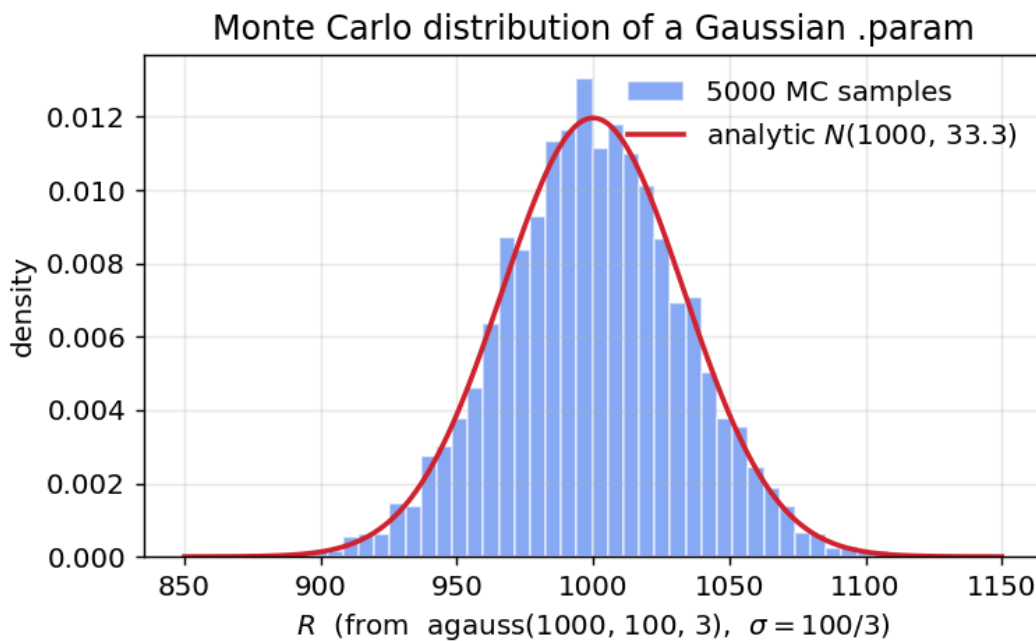
Capability	Command / syntax	Enhancement
Random parameters	agauss / gauss / aunif / unif / limit in .param	base
Deterministic seeding	setseed <s>; every loop command's -seed <s> (default 1, stated)	base, E-374, E-543
Ordinary Monte Carlo	the reset loop, or the alter loop	base (E-66)
Random draws without a re-source	the fast .param path, armed by montecarlo	E-346
Latin-Hypercube sampling	mcsample lhs <N> [seed <s>], montecarlo -lhs	E-149
High-sigma rare events	highsigma <N> -scale < λ > [-inflate <p>]... -metric <e> -max/-min	E-150, E-538
Process/mismatch correlations	mccorr <k> <matrix> + mvnorm(i)	E-151
Packaged yield	montecarlo <N> [-lhs] [-warm] -spec <e> -max/-min	E-151, E-188
Recording per sample	montecarlo ... -expr [name=<e>] → plot montecarlo<n>	E-552
Every hit of a condition, per sample	montecarlo ... -track "<track args>" → plot track<k>	E-577–584
The yield of a tracked quantity	-spec track1.value, -expr feeding a -track	E-609
Worst-case distance / MPFP	wcd -metric <e> -max/-min [-is <N>]	E-305
Design centring	optimize -dparam ... -center -samples <N> -spec ...	E-206
Model-declared statistics	(* std= / std_rel= / dist= / trunc= *) on a parameter + .option osdimc mcseed=<s>	E-530, E-554
Scoped importance weights	highsigma -inflate @owner[param]	E-538
Every draw of every run, to a file	.option savemc [=<file>], automc_save	E-610
Results onto the same row	writemc [name=<e>]..., montecarlo -writemc ...	E-611
Randomness inside the model	\$random, \$rdist_normal(seed, ...), ... in Verilog-A	E-10, E-527
Process corners	.lib / .include corner model sets	base

1. Random parameters

Statistical variation enters through the random functions in a `.param` expression. They are evaluated once per Monte Carlo sample, so each sample re-throws the dice:

function	draw	notes
<code>agauss(nom, avar, sig)</code>	Gaussian, mean <code>nom</code> , $\sigma = \text{avar}/\text{sig}$	"absolute": <code>avar</code> is the $\text{sig}\text{-}\sigma$ spread
<code>gauss(nom, rvar, sig)</code>	Gaussian, $\sigma = \text{nom} \cdot \text{rvar}/\text{sig}$	"relative" variation
<code>aunif(nom, avar)</code>	uniform on <code>[nom—avar, nom+avar]</code>	
<code>unif(nom, rvar)</code>	uniform on <code>[nom(1—rvar), nom(1+rvar)]</code>	
<code>limit(nom, avar)</code>	<code>nom ± avar</code> (a fair coin)	corner-style $\pm$
<code>mvnorm(i)</code>	component <code>i</code> of one correlated standard-normal draw	needs <code>mccorr</code> , §5

So `agauss(1000, 100, 3)` is a resistor with a nominal $1\text{ k}\Omega$ and a $3\text{-}\sigma$ spread of $100\ \Omega$, i.e. $\sigma = 33.3\ \Omega$. Drawing 5000 of them and histogramming recovers the Gaussian exactly:



The independence gotcha. Every *textual* occurrence of a random `{param}` draws independently — ngspice inlines the `.param`'s expression into each device line, so two devices written with the same `{rr}` get *different* values in one sample:

```
* two uses of one random .param are two draws
.param rr = agauss(1000, 100, 3)
V1 a 0 DC 1
R1 a 0 {rr}
R2 a 0 {rr}
.control
setseed 1
op
```

```

print @r1[resistance] @r2[resistance]
reset
op
print @r1[resistance] @r2[resistance]
.endc

@r1[resistance] = 1.017613e+03
@r2[resistance] = 1.049945e+03
@r1[resistance] = 1.022025e+03      <- the reset re-drew both
@r2[resistance] = 1.011230e+03

```

Matched/correlated devices need either the shared-.param idiom or the native correlation support in §5. The same inlining is why .option savemc (§8) names a draw by its slot — r1, r2 — and not by the .param.

A random function is refused where it cannot mean anything: in a Verilog-A parameter's *default* or *range* it is a compile error, not a crash ([E-545](#)).

2. The two Monte Carlo idioms

(a) The **reset** idiom — a random .param feeds a device/model, and each reset re-sources the deck (re-throwing the dice) and re-runs the analysis:

```

* reset-idiom Monte Carlo
.param rr = agauss(1000, 100, 3)
V1 a 0 DC 1
R1 a 0 {rr}
.control
  setseed 1
  let n = 500
  let iv = unitvec(n)
  let run = 0
  dowhile run < n
    reset          ; re-throws rr, re-runs
    op
    let iv[run] = -i(V1)
    let run = run + 1
  end
  print mean(iv) stddev(iv)
.endc
.end

mean(iv) = 9.996056e-04
stddev(iv) = 3.362374e-05

```

This is the idiom every command below builds on. It works with .model-card parameters (r={rr}) and OSDI/Verilog-A instances alike.

(b) The **alter** loop — control-language random vectors (sgauss(0), sunif(0)) assigned per run with alter; no netlist re-parse, so it is faster, but it does not benefit from mcsample/highsigma/mccorr, which target the .param idiom (that is where the well-defined per-sample boundary is).

setseed <s> makes either idiom bit-for-bit reproducible — including transient noise sources, which it did not seed before [E-374](#).

The fast path ([E-346](#)). montecarlo does not reset per sample: it captures every brace expression that draws from the RNG (agauss, gauss, unif, aunif, limit, mvnorm) and re-evaluates *those* in deck order,

pushing the values in place. The RNG stream is consumed exactly as a re-source would consume it, so the samples are bit-identical to the **reset** idiom — with `-lhs` too, since the sample boundary is still signalled to the sampler — and the banner says when it is on:

```
montecarlo: fast path armed (1 random value binding, no per-sample reset)
```

Warm start (E-188): `montecarlo ... -warm` reuses the previous sample's solution as the next sample's initial guess, which is what a slowly-moving parameter set wants.

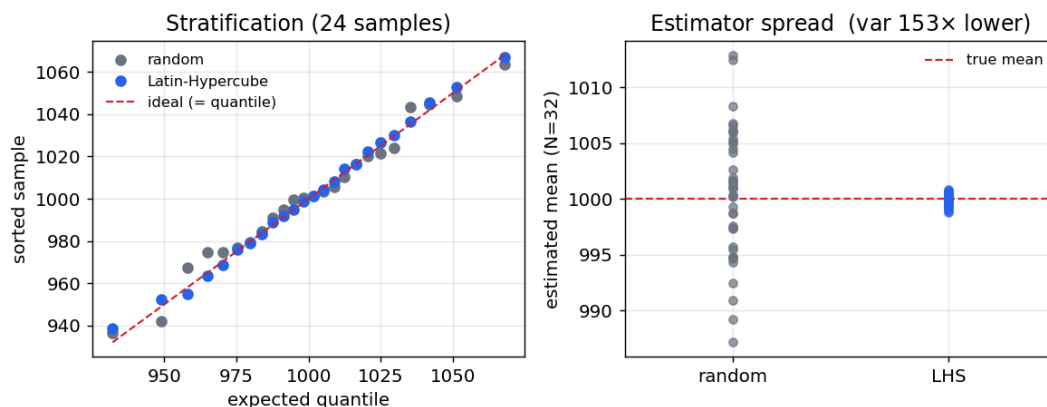
3. Latin-Hypercube sampling — **mcsample**

Plain Monte Carlo draws each parameter independently, so a modest run count clumps in some regions and leaves gaps in others, and the estimate converges only as $1/\sqrt{N}$. `mcsample lhs <N>` switches the `.param` draws to Latin-Hypercube sampling: each random dimension's range is split into N equal-probability strata and hit exactly once (with an independent stratum permutation per dimension; Gaussians are stratified in probability space through the inverse-normal CDF).

```
mcsample lhs <N> [seed <s>]    engage LHS for the next N reset-driven samples
```

```
mcsample random | off          revert to independent draws
```

The left panel shows the stratification (the LHS samples land on their quantiles; the random ones scatter); the right shows the payoff — the spread of the estimated mean over many trials collapses (here $\sim 100\times$ lower variance at the same N):



Usage is a one-line change to the idiom above:

```
.control
  mcsample lhs 500 seed 1          ; <-- the only change
  let run = 0
  dowhile run < 500
    reset
    op
    let iv[run] = -i(V1)
    let run = run + 1
  end
  print mean(iv) stddev(iv)        ; a much tighter estimate than plain MC
.endc
```

Monte Carlo sampling: Latin-Hypercube, $N = 500$, seed = 1.

```
mean(iv) = 1.001120e-03
```

```
stddev(iv) = 3.363422e-05
```

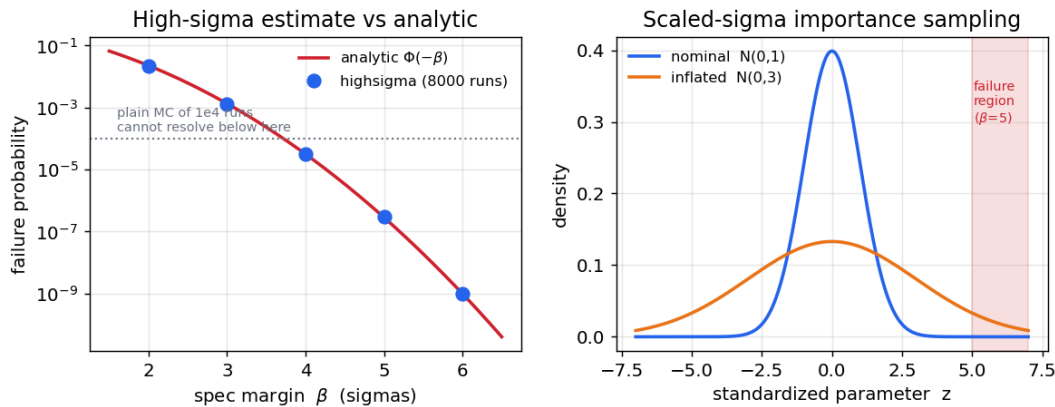
Use LHS whenever you want an accurate *whole-distribution* statistic (mean, stddev, a moderate quantile) for a given run budget.

4. High-sigma rare events — **highsigma**

For high-replication circuits (an SRAM cell instanced millions of times), the number that matters is a 4–6 sigma failure probability of $1e-7 \dots 1e-9$. Plain Monte Carlo would need $1e7 \dots 1e9$ runs just to see a handful of failures. **highsigma** estimates such probabilities with a few thousand runs by scaled-sigma importance sampling: it inflates every Gaussian .param's σ by a factor λ so the failure region is sampled often, then reweights each sample by the likelihood ratio $p_{\text{nominal}}/p_{\text{inflated}}$ to keep the estimate unbiased. It is direction-free — no gradient or worst-case-distance search.

```
highsigma <N> [-scale <lambda>] [-inflate <param>]... [-seed <s>] [-analysis <cmd>]
           -metric <expr> [-max <hi>] [-min <lo>]
```

The left panel shows the estimate tracking the analytic $\Phi(-\beta)$ from 2 to 6 sigma, far below what plain MC of the same budget can resolve; the right panel shows the mechanism — the inflated distribution reaches the tail the nominal one never does:



* probability that R exceeds a 4.5-sigma spec (1150 ohm); analytic $3.4e-6$

```
.param rr = agauss(1000, 100, 3)
```

```
V1 a 0 DC 1
```

```
R1 a 0 {rr}
```

```
.control
```

```
    highsigma 6000 -scale 3.0 -seed 1 -analysis op -metric -1/i(v1) -max 1150
```

```
    print highsigma_pfail highsigma_sigma highsigma_seed
```

```
.endc
```

```
.end
```

```
highsigma: 6000 samples, scale (sigma inflation) = 3, analysis 'op', fail if (-1/i(v1)) > 1150, seed 1
```

```
failures observed : 388 / 6000 (in the inflated sampling)
```

```
P(fail)           : 3.2349e-06 +/- 3.26e-07 (relative error 10.1%)
```

```
equivalent sigma  : 4.510 (one-sided, P = Phi(-sigma))
```

```
highsigma_pfail = 3.234899e-06
```

```
highsigma_sigma = 4.510425e+00
```

```
highsigma_seed = 1.000000e+00
```

The spec is `-max/-min` numeric limits rather than a `>/<` inside the metric, because a bare `>` in a control command is an I/O redirect. Results are also left in `highsigma_pfail`, `highsigma_relerr`, `highsigma_sigma`, `highsigma_nfail`, `highsigma_nfailed` (samples that did not solve), `highsigma_ess` and `highsigma_seed`.

4.1 Many dimensions: the weight collapses, **-inflate** scopes it

The importance weight is a product over every inflated dimension, so its variance grows exponentially with their number — and `-scale` inflates *every* Gaussian statistical parameter in the circuit, including

ones the metric cannot depend on. Under `.option osdimc` (§7) that count is not a modelling choice: a `(* type="instance" *)` mismatch parameter is one dimension per instance, so an ordinary deck arrives in the degenerate regime by itself. The estimator stays unbiased in expectation and useless in practice; the command therefore measures its own reliability — the effective sample size $ESS = (\Sigma w)^2 / \Sigma w^2$ — and says when it has none (E-537). The fix is not arithmetic but scope: `-inflate <param>` (E-538), repeatable, restricts the inflation (and so the weight) to the parameters the failure actually turns on — a bare name (`r`, wherever it occurs) or the accessor spelling `@owner[param]`, `*` allowed as the owner.

* the metric depends on `N1` alone; ten bystanders elsewhere each add a mismatch

↪ dimension

`.option osdimc`

`V1 1 0 DC 1`

`N1 1 2 mm`

`R1 2 0 1k`

`.model mm rstat r=1k ; rstat: (* std=25 *) r, (* type="instance",`
 ↪ `std=10 *) dr`

`V2 10 0 DC 1`

`N2 10 11 bys`

`... ; N2..N11 in a chain nothing measures`

`N11 19 0 bys`

`.model bys rstat r=1k`

`.control`

`pre_osdi rstat.osdi`

`highsigma 2000 -scale 2.5 -seed 1 -analysis op -metric v(2) -min 0.487`

`highsigma 2000 -scale 2.5 -seed 1 -analysis op -inflation @mm[r] -inflation @n1[dr]`

↪ `-metric v(2) -min 0.487`

`.endc`

`P(fail) : 6.6824e-03 +/- 6.31e-03 (relative error 94.5%)`

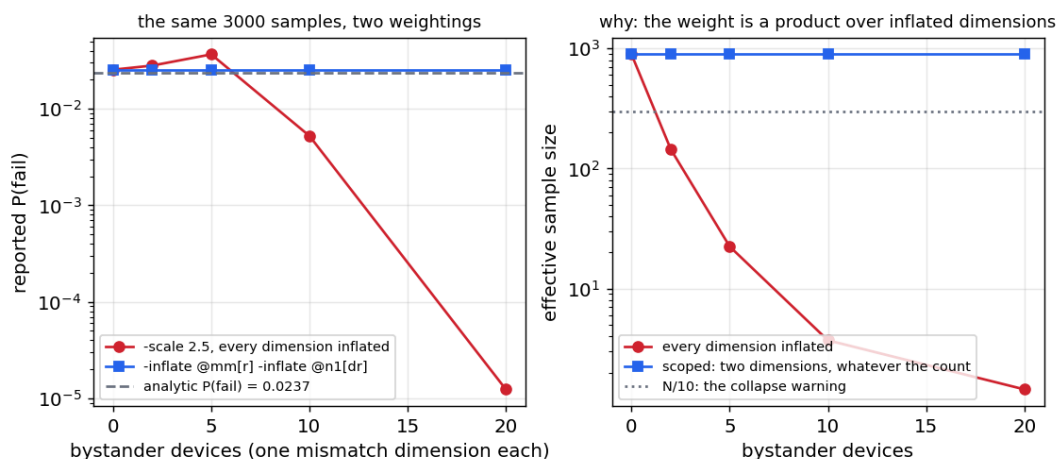
`NOTE : the importance weights have collapsed -- an effective sample size of 10.1 out of 2000. P(`
`-scale inflates EVERY (* std *) parameter of every device, including ones the metric cannot`
`inflation <param>`.`

`highsigma: 2000 samples, scale (sigma inflation) = 2.5 on the -inflation parameters only, analysis '`

`P(fail) : 2.5504e-02 +/- 2.55e-03 (relative error 10.0%)`

`equivalent sigma : 1.951 (one-sided, P = Phi(-sigma))`

The analytic answer is $\Phi(-53.4 / \sqrt{(25^2 + 10^2)}) = 0.0237$. The same 2000 samples were run both times — only the weighting differs — and the figure below sweeps the bystander count: the scoped estimate is *bit-identical* whatever the count, because the weight counts exactly the inflated dimensions.



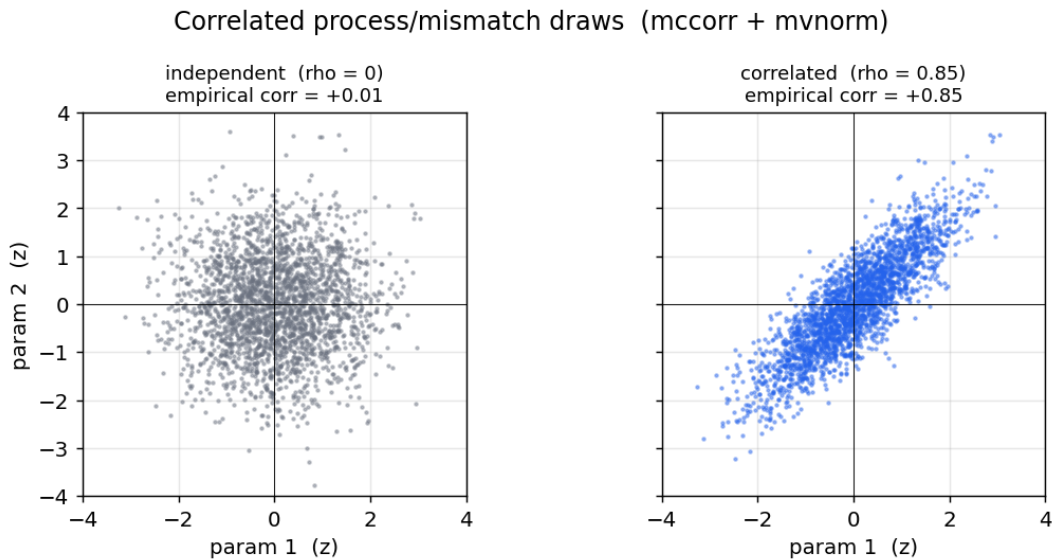
An `-inflate` naming nothing statistical is reported (the run then sampled the nominal spread), a malformed one is refused before the run. Two more honesty repairs from the same hunt: a weighted mean is clamped into `[0, 1]` and says when it was; an "equivalent sigma" at the boundary reads `n/a`, never `0.000` (which would mean $P = 0.5$).

5. Process/mismatch correlations — `mccorr` + `mvnorm`

Whether two devices' variation is *correlated* is often what decides the yield — but plain MC draws every `agauss` independently. `mccorr` registers a $k \times k$ correlation matrix (Cholesky-factored once, and rejected if not positive-definite), and the `.param` function `mvnorm(i)` returns the *i*-th component of one correlated standard-normal draw per sample ($y = L \cdot z$):

```
mccorr 2 1 0.85 0.85 1 ; rho = 0.85 between the two factors
.param r1 = 1000 + 100*mvnorm(1) ; r1, r2 now vary together
.param r2 = 1000 + 100*mvnorm(2)
```

`mvnorm(i)` returns a unit-variance standard normal (scale it in the `.param`); the matrix is a correlation matrix (unit diagonal). Drawing two parameters at $\rho=0$ vs $\rho=0.85$ shows the joint distribution tilt from a round blob to an elongated one:



The underlying *z*'s are drawn through the same sampler `mcsample/highsigma` use, so correlations compose with Latin-Hypercube and importance sampling automatically. The classic process + mismatch decomposition is just:

```
mccorr 1 1 ; one shared process factor
.param vth1 = 0.5 + 0.03*mvnorm(1) + 0.01*agauss(0,1,1) ; global + local
.param vth2 = 0.5 + 0.03*mvnorm(1) + 0.01*agauss(0,1,1) ; shares the process term
```

Here `mvnorm(1)` (the global process shift) is shared by both devices, while each `agauss(0,1,1)` is independent local mismatch — exactly the standard model. (With Verilog-A models the same split is one attribute: a model parameter is process, a `(* type="instance" *)` parameter is mismatch — §7.)

An index the matrix does not have is refused (MC hunt F4, 2026-09-04): with a $k \times k$ matrix registered, `mvnorm(0)`, `mvnorm(k+1)` or a fractional index is a `.param` error naming the range, where it used to fall through to an independent draw in silence. With no matrix registered `mvnorm(i)` draws independently by design — that is every deck's state at load, before its `.control` block has run `mccorr` — so `mccorr` itself reports an index the deck has already used beyond the matrix, and notes that the load-time draws are independent until a reset redraws them (the sampling commands do one per sample).

6. Yield — **montecarlo**

montecarlo packages the whole flow: run the samples, apply the pass/fail specs, report the yield with a confidence interval — or simply record a value per sample for later (`-expr`, E-552), every place a condition holds in each sample (`-track`, E-582), and the yield of a tracked quantity in the same command (E-609).

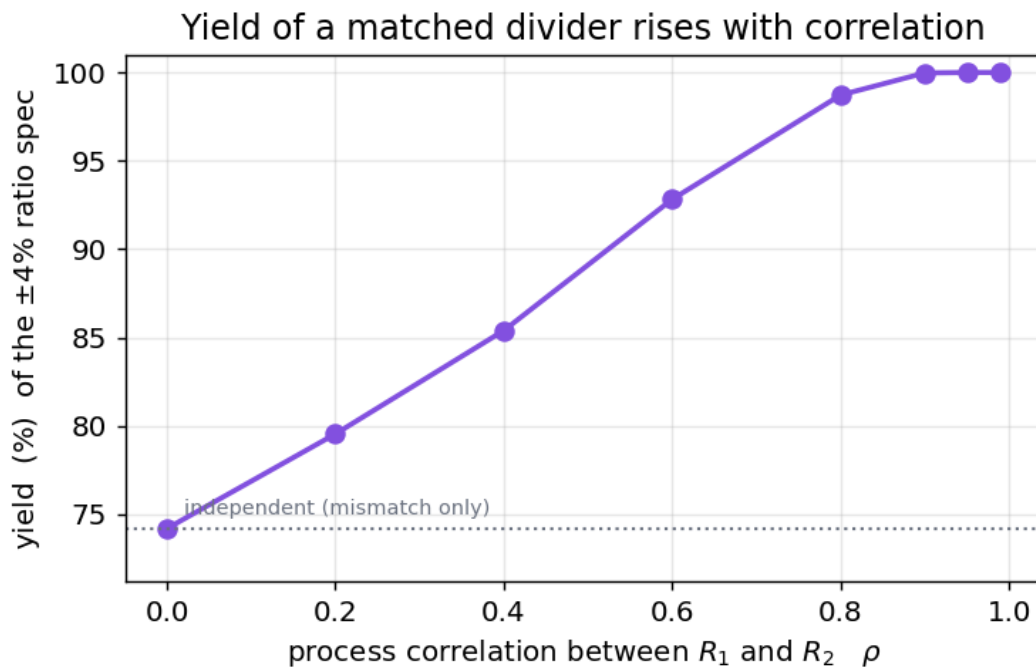
```
montecarlo <N> [-lhs] [-warm] [-seed <s>] [-analysis <cmd>]
              (-spec <metric> -max <hi>|-min <lo>)...
              (-expr [name=]<expression>)...
              (-track "<track arguments>")...
              [-writemc [name=]<expression>...]
```

A sample passes only if *every* spec's metric is within its `-max/-min` limits. It reports the yield with a Wilson 95% confidence interval and a per-spec violation count, and leaves `montecarlo_yield`, `montecarlo_npass`, `montecarlo_n`, `montecarlo_seed` for scripting. `-lhs` gives a much lower-variance yield estimate. A `-spec` is a judgement, so it needs a limit; one without `-max/-min` is refused with a pointer to `-expr`.

```
* a matched divider, +/-4% ratio spec: the yield depends on the correlation
.param r1 = 1000 + 50*mvnorm(1)
.param r2 = 1000 + 50*mvnorm(2)
V1 in 0 DC 1
R1 in out {r1}
R2 out 0 {r2}
.control
  mccorr 2 1 0.0 0.0 1
  montecarlo 4000 -lhs -seed 1 -analysis op -spec v(out) -max 0.52 -min 0.48
  mccorr 2 1 0.9 0.9 1
  montecarlo 4000 -lhs -seed 1 -analysis op -spec v(out) -max 0.52 -min 0.48
.endc
.end

montecarlo: 4000 Latin-Hypercube samples, analysis 'op', 1 spec, seed 1
  yield : 74.200% (2968 / 4000 pass)
  spec 1 (v(out)): 1032 violations
montecarlo: 4000 Latin-Hypercube samples, analysis 'op', 1 spec, seed 1
  yield : 99.975% (3999 / 4000 pass)
  spec 1 (v(out)): 1 violation
```

Because `mvnorm` correlations feed straight into **montecarlo**, the yield of a *matched* pair depends strongly on how correlated the two devices are — from ~74 % when independent to ~100 % when they track each other. Getting the correlation model wrong grossly misestimates the yield:



Seeds (E-543, E-559). Without `-seed` the netlist PRNG is re-seeded from 1, every time — so “run it again” returns the same samples, which is what a paired comparison across design changes wants and what a replication does not. The banner states it (`seed 1 (default)`), an un-seeded run notes that a rerun repeats it, and `-seed <n>` gives an independent replication. Under `.option osdimc` the model-declared draws are keyed by the same `-seed` and the sample number (§7.4), so a seeded run replays identically whatever ran before it.

A quoted or unresolvable spec is refused with the spec, sample and plot named — it used to score as 0 and report a confident 0 % or 100 % yield.

6.1 Recording without judging — `-expr`

`-expr [name=<expression>` evaluates the expression after every sample and records it, unjudged, into a plot of its own — `montecarlo1`, `montecarlo2`, ... one per invocation, with sample (1 ... N) as its scale, and named in `$montecarlo_plot`. With no `-spec` at all there is no yield: the command just runs the analysis N times and keeps what you asked for. A name is refused when the record plot already owns it — its scale sample, or a result such as `montecarlo_n` (E-557) — and when two `-expr` share it.

```
.param rr = agauss(1000, 100, 3)
V1 in 0 dc 1 ac 1
R1 in out {rr}
R2 out 0 1k
.control
  montecarlo 200 -seed 3 -analysis op -expr vo=v(out) -expr r=@r1[resistance]
  print mean(vo) stddev(vo) ; montecarlo1 is now current
  pyplot -hist vo

  montecarlo 50 -analysis "dc v1 0 1 0.01" -expr vo=v(out)
  plot vo ; 50 curves, one per sample
  print montecarlo1.vo[3] ; an earlier run is still there
.endc
```

What each `-expr` becomes:

the expression is	recorded as
a scalar per sample (<code>v(out)</code> after <code>op</code> , <code>vecmax(...)</code> , a device parameter)	an N-long vector on the sample scale
a waveform per sample (<code>v(out)</code> after a <code>dc/ac/tran</code> sweep, L points)	an $N \times L$ two-dimensional vector with the analysis scale (<code>v-sweep</code> , frequency, time) copied beside it — <code>plot</code> draws it as a family of N curves, <code>vo[k]</code> is sample k

A complex value (an `ac` output) is recorded as its magnitude. A sample that failed to simulate leaves `nan` in its row. A waveform whose point count differs between samples — an adaptive `tran` can do that — is not recorded and the command says so; reduce it to a scalar, or `linearize` it inside the `-analysis` command. An expression that gives the same value in every sample is noted, since it means nothing the deck draws reaches it. The name must be a plain identifier (`montecarlo1.<name>` has to be spellable); without one the vectors are `expr1`, `expr2`, ...

`-spec` and `-expr` combine: a `yield` run with `-expr` also records its values, so a `yield` and the distribution behind it come from the same samples.

6.2 Every place a condition holds, per sample — **-track**

`-expr` records one value, or one waveform, per sample. `track` (E-577) answers a different question — *every place a condition holds*: each ringing peak, each crossing of a threshold, each region above a limit — and it is a command, not an expression, so `-expr` cannot reach it and `-analysis` cannot run it (that flag runs exactly one command, the analysis). `-track "<track arguments>"` runs `track <arguments>` after every sample's analysis, quietly, and records the result into a plot of its own, **track<k>** — the shape of a track plot, stacked over the samples:

vector in <code>track<k></code>	holds
sample (the scale) hits	1 ... N the hit count per sample: 0 a miss, <code>nan</code> a sample that never solved
the scale of the analysis (time, frequency, <code>v_sweep</code> for a <code>dc</code> sweep), value (or <code>value1..N</code> , or the <code>-output</code> names), index, and a region's <code>x_out</code> and width — each with its type	an $L_{\max} \times N$ family, hit-major: row k (<code>time[k]</code>) is hit k of every sample on the sample scale, <code>nan</code> where a sample had fewer, $L_{\max}$ being the largest count any sample had — a <i>varying</i> count per sample is the usual case here, and the one <code>-expr</code> refuses; when no sample ever has more than one hit (a <code>-which first</code> selection, a single crossing) they are plain N-long vectors instead

The argument is one quoted word, because `track`'s own options begin with `-`; it is repeatable, and every `-track` gets its own plot, numbered from the first free `track<k>` name. `montecarlo<n>` stays the current plot afterwards — it holds the run's counts and the `-expr` vectors — so a record is reached as `track1.value` from there, or with `setplot $track_plot`; `$track_plot` names the last record made and `$track_hits` counts the samples that had a hit. `-track` combines with `-spec` and `-expr` in the same run. The per-sample track plots are destroyed as they are copied. A miss is silent and records 0; a real error in the track arguments — an expression that does not evaluate, an unknown option — stops the run on its first sample with the message once, and records nothing.

```
.param rr = agauss(20, 30, 3)
V1 in 0 pulse(0 1 0 1n 1n 1m 2m)
```

```

R1 in a {rr}
L1 a out 1m
C1 out 0 1u
.control
  montecarlo 1000 -seed 3 -analysis "tran 2u 1m" -track "v(out) -spec localmax
  ↪ -prominence 20m" -expr r=@r1[resistance]
  print mean(track1.hits) ; ringing peaks per sample
  plot track1.value[0] vs r ; the first peak's height against the
  ↪ resistance
  setplot $track_plot
  pyplot -hist time[1] ; where the second peak lands (nan where
  ↪ there was none)
.endc

```

value[0] is row 0 of the family — the first hit of every sample, N long — and time[1] the second, nan where a sample had fewer; time[k][i] is sample i's k-th hit; plot time draws Lmax curves against sample. The orientation is the opposite of the -expr waveform families above, where vo[k] is sample k's curve, because the question about hits is "where did the second peak land across the samples", not "what did sample k do". -expr beside it keeps the per-sample cause, here the resistance, on the same sample scale in montecarlo<n>, so a hit count or a peak position can be plotted against the parameter that drove it. A crossing needs a previous sample on the other side, and transition/slew are unity in a small-signal analysis (E-587).

6.3 The yield of a tracked quantity — -spec reads the track, -expr feeds it

Before E-609 a -spec was evaluated on the analysis plot, before the tracks ran, so nothing in it could reach a track's result. Now the tracks run first and each sample's track plot is kept until the specs and exprs have read it:

- **track<k>.<vector>** in a -spec or -expr names the k-th -track of the command, for the sample being judged — track1.value, track2.x_out, track1.value[0] (the first of several hits; a bare track1.value judges the last), and track1.hits, the hit count as a number, 0 on a miss. A miss under a -spec is a violation, counted apart in the report ("... 24 violations (23 of them samples whose track had no hit to judge)"); -spec track1.hits -max 0 is the yield of "nothing there".
- An -expr feeds a -track or a -spec. An -expr that does not read a track is evaluated *before* the tracks and defined as a vector of that name in the sample's plot, so a track's expression and a spec may use it by name.

* -expr feeds a -track whose -output a -spec then judges; a second -expr records
 ↪ the cause

```

.param rr = agauss(1000, 100, 3)
V1 in 0 dc 1
R1 in out {rr}
R2 out 0 1k
.control
  montecarlo 200 -seed 3 -analysis "dc v1 0 5 0.01" -expr q=v(out)*2 -track "q
  ↪ -spec globalmax -output pk" -spec "track1.pk" -min 4.95 -expr r=@r1[resistance]
  print mean(montecarlo1.r) montecarlo_yield
  print track1.pk[0] montecarlo1.r[0]
.endc

```

montecarlo: fast path armed (1 random value binding, no per-sample reset)

montecarlo: -track "q -spec globalmax -output pk": a hit in 200 of 200 samples, never more than one
 - plot track1: sample, hits, v_sweep, pk, index (200 long)

montecarlo: 2 expressions over 200 samples recorded into plot 'montecarlo1' (now current)

```

yield : 71.500% (143 / 200 pass)
spec 1 (track1.pk): 57 violations
mean(montecarlo1.r) = 1.000704e+03
track1.pk[0] = 4.962493e+00
montecarlo1.r[0] = 1.015116e+03

```

q peaks at $10k/(rr+1k)$, so the spec passes when $rr < 1020 \Omega - 0.6 \sigma$ above the mean, ~73 % analytically. A metric may mix the two plots ($track1.v_sweep / \text{maximum}(v(out))$); a `track<k>` beyond the `-track` flags given is refused at parse time.

7. Model-declared statistics — `.option osdimc`

Everything above puts the statistics in the *netlist*: a `.param` draws, a device reads it. A Verilog-A model can instead declare its own variability where the parameter lives, and the simulator runs the whole Monte-Carlo loop (E-530): no reset, no netlist re-expansion, no `gauss()` in the deck, and process-versus-mismatch falls out of the model/instance split the language already has.

7.1 The attributes

```

(* std=25.0 *)                parameter real r = 1000.0 from (0:inf); //
↪ gauss, sigma 25
(* dist="uniform", std=2e-4 *) parameter real g = 1e-3;    // uniform: std is
↪ the HALF-WIDTH
(* std_rel=0.05 *)            parameter real k = 2.0;      // sigma = 5 % of
↪ the resolved nominal
(* type="instance", std=10.0 *) parameter real dr = 0.0;   // per-device
↪ mismatch
(* dist="lognormal", std_rel=0.3 *) parameter real is = 1e-15 from (0:inf); //
↪ never crosses zero
(* std=25.0, trunc=2.0 *)     parameter real rs = 1000.0; // a gauss
↪ confined to +-2 sigma
(* dist="tgauss", std_rel=0.05 *) parameter real kt = 2.0; // gauss with
↪ trunc=3

```

attribute	meaning
<code>std=<σ></code>	absolute standard deviation (uniform: the interval's half-width)
<code>std_rel=<σ></code>	σ relative to the nominal resolved at setup — so it follows a value the deck sets
<code>dist="gauss" (default) / "uniform"</code>	the shape
<code>dist="lognormal" (alias lnorm)</code>	$\text{nominal} \cdot \exp(s \cdot z)$; <code>std_rel</code> is the sigma of the logarithm, an absolute std is converted at the nominal (E-554)
<code>trunc=<n></code>	the Gaussian coordinate confined to $ z \leq n$ by deterministic rejection: a draw inside the window is exactly the untruncated draw; composes with <code>gauss</code> and <code>lognormal</code>
<code>dist="tgauss"</code>	gauss with <code>trunc=3</code>
<code>type="instance"</code>	composes with all of the above: an instance parameter draws per device (mismatch), a model parameter once per <code>.model</code> card (process)

A negative or non-literal sigma and `std` beside `std_rel` are located compile errors; an unknown distribution, statistics on a non-real parameter or a `localparam`, `dist` without a sigma, or `trunc` on a uniform are warnings. The statistics ride an optional OSDI side-table, so an object compiled without them loads

unchanged and one with them loads in an older simulator, without the statistics. A quoted number (`std_rel="0.05", trunc="2.5"`) is accepted.

7.2 Every run is a trial

With `.option osdimc` (alias `automc`) set, every run-class command — `op`, `tran`, `dc`, `ac`, `run`, ... — starts a fresh *trial*: each declared parameter is written `nominal + draw` through the ordinary parameter setter, the `alter` path, so node-collapse decisions and everything else downstream re-derive normally. The first run after sourcing is the nominal baseline — a parameter the deck never set has its default resolved by the model's own setup, so nominals are only knowable after one pass — and draws begin with the second run.

* model-declared statistics: the simulator draws, every run is a trial

```
.option osdimc mcseed=42
V1 in 0 DC 1
N1 in mid rm
N2 mid 0 rm
.model rm rstat r=1k          ; rstat: (* std=25 *) r, (* type="instance", std=10
  ↪ *) dr
.control
pre_osdi rstat.osdi
repeat 4
  op
  print @rm[r] @n1[dr] @n2[dr] v(mid)
end
.endc

trial 1: @rm[r]=1.000000e+03 @n1[dr]=0.000000e+00 @n2[dr]=0.000000e+00 v(mid)=5.000000e-
01
trial 2: @rm[r]=1.022885e+03 @n1[dr]=1.004356e+01 @n2[dr]=8.686182e+00 v(mid)=4.996713e-
01
trial 3: @rm[r]=1.008466e+03 @n1[dr]=1.742020e+00 @n2[dr]=5.958456e+00 v(mid)=5.010413e-
01
trial 4: @rm[r]=1.002764e+03 @n1[dr]=-1.05375e+01 @n2[dr]=2.809054e+00 v(mid)=5.033403e-
01
```

Trial 1 is the baseline. From trial 2 the model parameter `r` draws once per card — `N1` and `N2` share `rm`, so they move in lockstep (process) — while the instance parameter `dr` draws per device (mismatch): the divider leaves 0.5 only because the two `dr` differ. The drawn values are readable per trial through the ordinary `@inst[param] / @model[param]` channel, and `.option osdimc_verbose` prints every draw as it is made.

7.3 **mcseed**: draws are pure functions, not a stream

There is no random-number *stream* on the model side. A draw is a hash (`splitmix64` into Box–Muller) of four things — `mcseed`, the trial number, the owner (model-card or instance name) and the parameter id. Consequences:

- bit-for-bit reproducible: the same deck and `mcseed` give the same values in any process, in any order; trial `N` is re-runnable in isolation; resume never redraws. Trial 2 of the deck above draws `r = 1022.885` — and so does trial 2 of any other command sequence with the same seed (§7.4 shows it).
- `.option mcseed=<int>` swaps the whole ensemble; the default is 1.
- **alter/altermod** recentre: a stored value becomes the new nominal, and draws stay `nominal + 6` — never a random walk. A statistical parameter the deck never gave, whose default reads the

written one — a child's binding in a hierarchy (leaf `#(.r(rl)) c1`, flattened to `c1__r`), a plain parameter `real rb = rl`, an instance default from an instance parameter — is re-resolved by the next setup and its draws sit on the new value (E-614); `altermod tm rl=500` moves `c1__r` from `1000 +  $\delta$` to `500 +  $\delta$` , the same δ .

- turning the option off (`unset osdimc`, or `.option noosdimc / noautomc`) restores every drawn parameter to its nominal on the next run — a value the user gave by `alter`, not the deck's default (E-614); a re-source invalidates the table.

A seed that is not an integer is truncated and said; one that is not a number is refused and said, each once (E-572):

Warning: `.option mcseed=1.5` is not an integer; using 1

Warning: `.option mcseed=abc` is not a number; using the default seed 1

7.4 The trial policy for loop commands

A per-run draw is right for a lone `op` and wrong at both ends of the loop spectrum, so the loop commands take a position (E-535):

command	trials it consumes
<code>op</code> , <code>tran</code> , <code>dc</code> , <code>ac</code> , ...	one each — a <code>.dc</code> sweep or a <code>tran</code> is one sample, not N stitched ones
<code>sweep</code> , <code>optimize</code> , <code>wcd</code> , <code>loadpull</code>	one for the whole loop (a swept curve is one sample; an optimiser fits a fixed objective)
<code>montecarlo</code> , <code>highsigma</code> , <code>wcd -is</code>	one per sample, keyed by the command's <code>-seed</code> and the sample number instead of the trial counter (E-559) — so a seeded loop command replays identically whatever ran before it, and sample 1 is already a draw
a user <code>reset</code>	restarts at the baseline — a command's own internal reset preserves the sequence, yours does not

* the trial policy: a sweep is one trial; `altermod` recentres; `off` restores

```
.option osdimc mcseed=42 osdimc_verbose
```

```
V1 in 0 DC 1
```

```
N1 in 0 rm
```

```
.model rm rstat r=1k
```

```
.control
```

```
pre_osdi rstat.osdi
```

```
op ; trial 1, the baseline
```

```
dc v1 0 1 0.25 ; trial 2: one draw for the whole sweep
```

```
print @rm[r]
```

```
altermod rm r=2000 ; recentre
```

```
op ; trial 3 draws around 2000
```

```
print @rm[r]
```

```
unset osdimc
```

```
op ; nominal again
```

```
print @rm[r]
```

```
.endc
```

```
osdimc: trial 2: n1:dr = 10.0436 (nominal 0)
```

```
osdimc: trial 2: rm:r = 1022.89 (nominal 1000)
```

```
@rm[r] = 1.022885e+03
```

```
osdimc: trial 3: n1:dr = 1.74202 (nominal 0)
```

```
osdimc: trial 3: rm:r = 2008.47 (nominal 2000)
@rm[r] = 2.008466e+03
@rm[r] = 2.000000e+03
```

The deltas are those of trials 2 and 3 in §7.2 (+22.885, +8.466): same seed, same trial, same owner — same draw, whatever the command.

Under `montecarlo`, `-seed` pins the model-side draws as it pins the netlist ones, and the verbose line shows the pair:

```
.control
pre_osdi rstat.osdi
op
op
montecarlo 3 -seed 1 -analysis op -expr r=@rm[r] -expr d1=@n1[dr]
op
montecarlo 3 -seed 1 -analysis op -expr r=@rm[r]          ; a different history, the
↪ same samples
montecarlo 3 -seed 9 -analysis op -expr r=@rm[r]          ; a different ensemble
set osdimc_verbose
montecarlo 2 -seed 9 -analysis op -expr r=@rm[r]
.endc

Index  montecarlo1.r  montecarlo1.d1
0      9.984373e+02   -1.61130e+01
1      1.018678e+03   -5.96126e+00
2      9.748142e+02   -8.64998e+00
Index  montecarlo2.r
0      9.984373e+02           <- identical to montecarlo1
1      1.018678e+03
2      9.748142e+02
Index  montecarlo3.r
0      9.993001e+02
1      1.012530e+03
2      9.642309e+02
osdimc: trial 13: n2:dr = -18.934 (nominal 0) [sample 1 of -seed 9]
osdimc: trial 13: n1:dr = -2.1765 (nominal 0) [sample 1 of -seed 9]
osdimc: trial 13: rm:r = 999.3 (nominal 1000) [sample 1 of -seed 9]
```

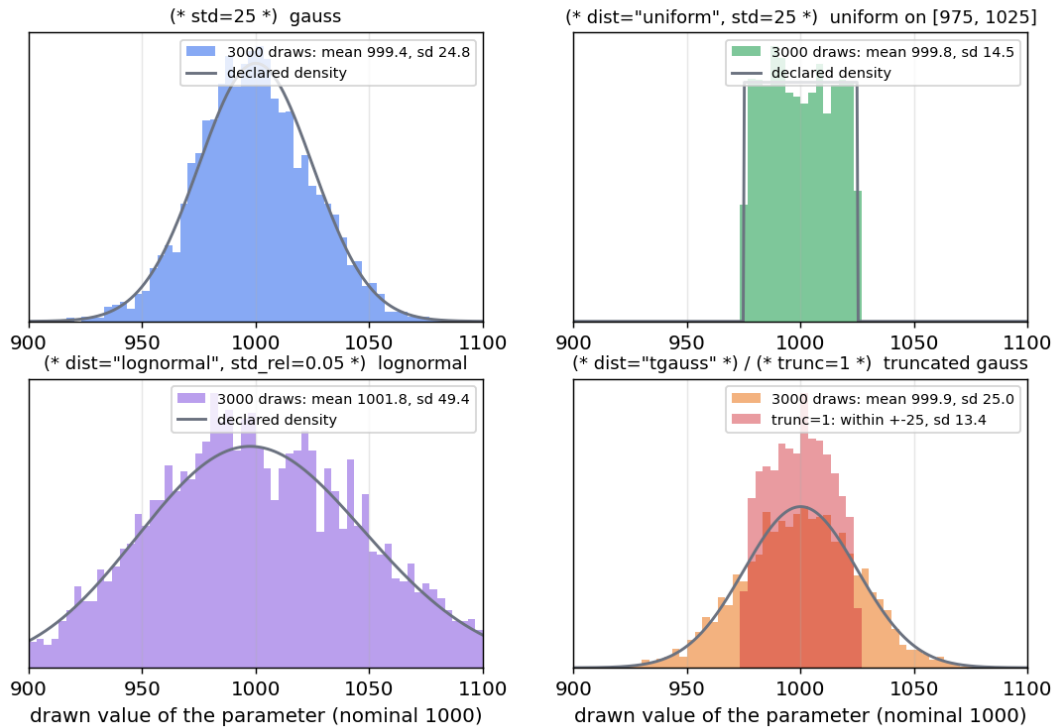
The **reset**-loop gotcha. Because a user `reset` restarts the sequence at the baseline, a hand-written `repeat ... reset ... tran` loop under `.option osdimc` redraws the *netlist* `.params` only — the model-declared parameters sit at their nominals on every pass (§8 shows the rows). Use `repeat N` with the analysis and no `reset` when only the model side varies, or `montecarlo`, whose internal per-sample reset preserves the sequence, when both must.

Two more rules. Machine writes — a `.dc` parameter sweep, the sweep command's points, a sens perturbation — pin their parameter for the run and are never overwritten by a draw (E-535). And a parameter the model tests with `$param_given` and the deck never gave is not drawn: the draw would switch the model to its "given" branch instead of varying it (BSIM4 derives `toxp` from `toxe` unless `toxp` is given), so the simulator says so once and leaves it alone (E-555).

7.5 The shapes, measured

The five shapes, 3000 draws each straight off a device through `montecarlo -expr:`

.option osdimc -- the simulator draws what the Verilog-A declares



* one parameter per shape, 400 draws each, measured

.option osdimc mcseed=3

V1 a 0 DC 1

N1 a 0 dm

.model dm rdlist

.control

pre_osdi rdlist.osdi

montecarlo 400 -analysis op -expr rg=@dm[rg] -expr ru=@dm[ru] -expr rl=@dm[rl]

↪ -expr rt=@dm[rt] -expr r1=@dm[r1]

print mean(rg) stddev(rg)

print minimum(ru) maximum(ru)

print minimum(rl) mean(ln(rl/1000)) stddev(ln(rl/1000))

print minimum(rt) maximum(rt) stddev(rt)

print minimum(r1) maximum(r1) stddev(r1)

.endc

mean(rg)=9.973730e+02 stddev(rg)=2.573553e+01

; (* std=25 *)

minimum(ru)=9.750660e+02 maximum(ru)=1.024806e+03

; uniform on [975, 1025]

minimum(rl)=8.643095e+02 mean(ln(rl/1000))=-5.48523e-04 stddev(ln(rl/1000))=4.629092e-02 ; lognormal, s = 0.05

minimum(rt)=9.316744e+02 maximum(rt)=1.063867e+03 stddev(rt)=2.560626e+01 ; tgauss: within +-75

minimum(r1)=9.751889e+02 maximum(r1)=1.024977e+03 stddev(r1)=1.382834e+01 ; trunc=1: within +-25, sd 0.55 sigma

Why the shapes matter: a gauss has no regard for a from (0:inf) range. A draw that violates the parameter's range fails that trial with the model's own located range error, exactly as the same alter would (the descriptor exports no ranges), and the loop commands drop and count the trial — under highsigma -scale in exactly the tail being measured. A lognormal cannot cross zero and a truncation cannot leave its window, so size sigmas on a bounded parameter with one of them. Both inflate under highsigma -scale with the matching importance weight (a truncated tail read ~30 % low without its

normaliser), and both take a wcd walk coordinate, clamped at the truncation.

7.6 With **highsigma** and **wcd**

The model-declared parameters are dimensions of both rare-event commands. For **highsigma** `-scale` each inflated one is a factor of the importance weight — which is why §4.1's `-inflate` exists, and why the figure there is drawn with `(* type="instance" *)` bystanders. For **wcd** every `gauss (* std *)` parameter of every model card and instance is a coordinate (the banner counts them, after the netlist ones); a uniform holds at its nominal, a lognormal takes its coordinate in the log domain, a truncated one is clamped at $\pm n$ and **wcd** says so when the boundary lies beyond the window:

* worst-case distance with model-declared statistics as the dimensions

```
.option osdimc
```

```
V1 in 0 DC 1
```

```
N1 in out mm
```

```
R1 out 0 1k
```

```
.model mm rstat r=1k
```

```
.control
```

```
pre_osdi rstat.osdi
```

```
op
```

```
wcd -analysis op -metric v(out) -min 0.487
```

```
wcd -analysis op -metric v(out) -min 0.487 -is 2000 -seed 1
```

```
.endc
```

```
wcd: 2 statistical dimensions (0 netlist .param, 2 model-declared), analysis 'op', fail if (v(out)
```

```
  worst-case distance : beta = 1.9828 sigma
```

```
  P(fail), first-order: 2.369579e-02 (= Phi(-beta))
```

```
  MPFP (standardised normal coordinates):
```

```
    u0=+0.7364 u1=+1.8410
```

```
  refining with 2000 mean-shift importance samples centred on the MPFP, seed 1...
```

```
  failures seen      : 987 / 2000 (in the shifted sampling)
```

```
  P(fail), mean-shift : 2.306361e-02 +/- 7.90e-04 (relative error 3.4%)
```

```
  equivalent sigma   : 1.994
```

u_0 is dr (σ 10), u_1 is r (σ 25): the boundary $r + dr = 1053.4$ is reached most probably by moving the wider parameter more — $0.7364 \cdot 10 + 1.8410 \cdot 25 = 53.4$. The three estimates of the same event agree: FORM 0.0237, mean-shift 0.0231, scoped **highsigma** 0.0255 ± 0.0026 (§4.1).

8. Recording every draw — **.option savemc,writemc**

A run with statistics produced results but no record of the draws behind them; reconstructing a sample's parameter set meant printing every `@dev [param]` in a loop. **.option savemc** (E-610) records, for every run-class command, one row with the value in force of every parameter with statistics:

column	what it is
trial, analysis, status	1, 2, ...; op/tran/...; ok, or failed for a run that did not solve (the draw happened)
r1,m1:w, rmod:r_ ohm	a device slot whose value draws (a random .param is inlined into each use, so each use is its own draw, named by the slot), a .model card's key={...}
x1.p	a subcircuit call's own drawn value (empty on montecarlo's fast path, which does not re-derive it)
r.x1.r1	a slot inside a subcircuit that reads a drawn symbol

column	what it is
@rm[r], @n1[dr]	under .option osdimc, every OSDI parameter with declared statistics, read off the device when the row is made — process ones by model, mismatch ones by instance

Columns are fixed by the first row and grow if a later row brings a new name. The file is mcparams_<YYYYMMDD>_<HHMMSS>.<ext> beside the netlist (the working directory for a deck not read from a file), made unique within a second; savemc=csv (default), savemc=txt (tab-separated), savemc=excel (a genuine .xlsx, rewritten every 25 rows and at exit; a model parameter's name — rm:r, @rm[r] — is bold in the header and an instance parameter's regular, E-617) or savemc=<name>.<csv|txt|xlsx> for a named file (an absolute path is taken as is; the name keeps its case and its bytes, and a quoted name its spaces — E-612; its directories are made, and a name that cannot be opened is said with the reason and the rows go to the dated default beside the netlist instead — E-613). One file per deck: a reset continues it, every montecarlo sample is a row, nosavemc turns it off; a different deck sourced in the same session with the same fixed name gets <stem>_2.<ext> and a note saying whose rows the first file holds (E-615). **.option automc_save** (alias osdimc_save, same values) records the OSDI draws only.

```
* every parameter with statistics, one row per run: .option savemc
.option savemc=draws.csv osdimc mcseed=7
.param rr = agauss(20, 30, 3)
.subckt load a b c=1u
C1 a b {c}
.ends
V1 in 0 pulse(0 1 0 1n 1n 1m 2m)
R1 in a {rr}
L1 a out 1m
x1 out 0 load c={gauss(1u, 0.1, 1)}
N1 out 0 rm
N2 out 0 rm
.model rm rstat r=1k
.control
pre_osdi rstat.osdi
op
montecarlo 3 -seed 3 -analysis "tran 2u 1m" -expr pk=maximum(v(out))
.endc
```

Note: savemc: recording the 6 parameters with statistics, one row per analysis run, to ./draws.csv
draws.csv:

```
trial,analysis,status,r1,x1.c,c.x1.c1,@n2[dr],@n1[dr],@rm[r]
1,op,ok,1.53917313235,1.03221785959e-06,1.03221785959e-06,0,0,1000
2,tran,ok,24.5348205054,,9.04013857186e-07,4.51220199384,0.91962064256,1018.16902622
3,tran,ok,21.1828413965,,1.18082157571e-06,13.3052107233,-2.23727670569,969.632099409
4,tran,ok,15.9771796274,,1.02835122004e-06,-8.38219382059,7.05127653696,1006.916445
```

Row 1 is the op — the osdimc baseline (0, 0, 1000) — and rows 2–4 the three samples: r1 and c.x1.c1 re-drawn per sample on the fast path, x1.c empty there, the model's r shared by N1 and N2, their dr independent.

One parameter on both channels. .model rm rstat r={agauss(1000,300,3)} with (* std=25 *) on r: the netlist's draw is the sample's nominal and the model's δ sits on it, so the row reads @rm[r] = rm:r + δ — on montecarlo's fast path as on its re-source path (E-616; the fast path used to apply δ over the *first* sample's draw on every sample). Which loop varies what: montecarlo varies both channels; a repeat ... reset ... op loop varies the netlist channel only — a user reset restarts the trial sequence,

so the model channel is its baseline on every pass (E-535) — and a repeat ... op loop without the reset varies the model channel only.

8.1 Results onto the same row — **writemc**

The row is written the moment the run ends; what the `.control` block computes from the run — a peak, a meas result, a track's hit count — comes after. **writemc [name=<expression> ...** (E-611) puts scalar values onto the last run's row (the csv/txt line is rewritten in place), and **montecarlo ... -writemc ...** does the same per sample, evaluated after the tracks, specs and exprs — so an `-expr` name, a `track<k>`, `<vector>` and any scalar expression may be listed:

* writemc: what the run produced, on the row of the draws behind it

```
.option savemc=results.csv osdimc mcseed=7
```

```
.param rr = agauss(20, 30, 3)
```

```
V1 in 0 pulse(0 1 0 1n 1n 1m 2m)
```

```
R1 in a {rr}
```

```
L1 a out 1m
```

```
C1 out 0 1u
```

```
N1 out 0 rm
```

```
.model rm rstat r=1k
```

```
.control
```

```
pre_osdi rstat.osdi
```

```
montecarlo 3 -seed 3 -analysis "tran 2u 1m" -expr pk=maximum(v(out)) -track "v(out)
```

```
↳ -spec localmax -prominence 20m" -writemc pk npk=track1.hits tpk=track1.time[0]
```

```
↳ over=maximum(v(out))-1
```

```
repeat 2
```

```
  reset
```

```
  tran 2u 1m
```

```
  let pk = maximum(v(out))
```

```
  meas tran tr trig v(out) val=0.1 rise=1 targ v(out) val=0.9 rise=1
```

```
  writemc pk tr q=pk*2
```

```
end
```

```
.endc
```

results.csv:

```
trial,analysis,status,r1,@n1[dr],@rm[r],pk,npk,tpk,over,tr,q
```

```
1,tran,ok,24.5348205054,0.91962064256,1018.16902622,1.22587470677,2,0.000107080689285,0.225874
```

```
2,tran,ok,10.4013857186,-2.23727670569,969.632099409,1.54646928156,4,0.000100490845987,0.546469
```

```
3,tran,ok,21.1828413965,7.05127653696,1006.916445,1.28611461892,2,0.000104868548536,0.28611461
```

```
4,tran,ok,38.0821575706,0.1000,1.05111554647,,,,6.165155e-05,2.10223109293
```

```
5,tran,ok,15.5321707455,0.1000,1.3976247175,,,,4.074237e-05,2.795249435
```

Rows 1–3 are the montecarlo samples with the four `-writemc` values; rows 4–5 the hand loop with its own three — a column a row never gets stays empty. Note `@n1[dr] = 0, @rm[r] = 1000` on rows 4–5: the reset restarted the model-side sequence at its baseline (§7.4), while `r1` kept drawing. A `writemc` item must be a scalar (a waveform is refused with its point count — reduce or index it; element `[0]` of a one-point vector is itself); a name that is a draw's (`r1`) lands as `r1*` beside the draw; with `savemc` off the command says so once and does nothing, so one loop runs with and without recording.

8.2 From a schematic front end

The same machinery runs under a host that loads `libngspice` and hands the circuit over as text — KiCad's simulator, driven through its GUI in the KiCad/example6 project of the companion software—builds tree (E-611 records the run). Three things differ from a netlist run: give `savemc` an absolute

path (the host sets no netlist directory); a `.control` block runs when the circuit is loaded, *before* the host's own analysis, so a `writemc` meant for that run waits behind `set controls wait`; and every root-sheet net is spelled `/name`, which `v(/mid)/v(/in)` now parses everywhere an expression is evaluated. A `.model` card whose parameter draws — `.model rmod va_res R_ohm={agauss(1k,50,3)}`, the natural place for process variation — is a `rmod:r_ohm` column on both the fast path and a plain run.

9. Randomness inside the model — `$random`, `$rdist_*`

Verilog-AMS's random system functions — `$random`, `$arandom`, `$rdist_uniform`, `$rdist_normal`, `$rdist_exponential`, `$rdist_poisson`, `$rdist_chi_square`, `$rdist_t`, `$rdist_erlang` and their integer `$dist_*` forms — compile and run (E-10, audited against the LRM in E-527). The LRM's in-out seed that advances on every call cannot work in a Newton loop (the value would change between iterations and the residual never converge), so each call is a pure function of (seed value, call-site salt, arguments) with no write-back: reproducible, stable across the solve, independent per call site — and per-instance when the seed is an instance parameter, which is the dominant use, mismatch:

```
module rseed(p, n);
  inout p, n; electrical p, n;
  (* type="instance" *) parameter integer seed = 1;
  parameter real r = 1000.0;
  (* desc="drawn resistance" *) real rr;
analog begin
  rr = r * (1.0 + 0.05 * $rdist_normal(seed, 0.0, 1.0)); // 5 % mismatch, per
  ↪ instance
  I(p, n) <+ V(p, n) / rr;
end
endmodule

V1 a 0 DC 1
N1 a 0 sm seed=1
N2 a 0 sm seed=2
N3 a 0 sm seed=2
.model sm rseed r=1k
.control
pre_osdi rseed.osdi
op
print @n1[rr] @n2[rr] @n3[rr]
setseed 77
op
print @n1[rr] @n2[rr] @n3[rr]
.endc

@n1[rr] = 1.027611e+03 @n2[rr] = 1.074738e+03 @n3[rr] = 1.074738e+03
@n1[rr] = 1.027611e+03 @n2[rr] = 1.074738e+03 @n3[rr] = 1.074738e+03
```

Two instances with the same seed draw the same value, a different seed a different one, and `setseed` — the *netlist* PRNG — does not touch them. This is the third statistical channel, and the least automated: the deck (or a `.param` with `agauss`, or `.option osdimc` on the seed parameter itself) has to vary the seed. For a parameter that should simply vary from run to run, the attributes of §7 are the better tool; `$rdist_*` is for randomness the *model* owns — a draw inside an equation, a distribution the attributes do not offer.

10. Corners, centring and worst-case distance

Process corners (TT / FF / SS / ...) are the ordinary ngspice `.lib/.include` mechanism — a corner model set is selected by a `.lib "models.lib" ff` line. Corners compose with all of the above: load a corner, then run `montecarlo` at that corner to get the corner's yield, or loop the corners around the MC. A typical production flow is therefore:

1. `.lib` in a process corner (or loop over corners);
2. declare correlated process/mismatch with `mccorr + mvnorm`, or let the models declare theirs and set `.option osdimc`;
3. `montecarlo ... -lhs` for the yield (or `highsigma ...` for a rare-event spec), with `.option savemc` recording the draws behind every row.

Design centring (E-206): `optimize -center` moves the nominal design point to maximise the fraction of the statistical population that meets spec, rather than optimising the nominal response alone. The outer optimiser searches the design knobs; at each candidate the inner loop runs *N* Monte-Carlo samples around the current centre, evaluates every spec, and maximises the worst-case Cpk — continuous where raw yield is a step function. A fixed inner seed gives every candidate the same draws (common random numbers); under `.option osdimc` the inner loop draws too, and every candidate replays the same window of trials — the counter is rewound to a checkpoint per candidate — so the objective samples the model's statistics and is still deterministic across candidates (E-536).

```
* design centring: move the nominal so most of the population meets spec
.param xc = 4.0
.param vo = agauss(xc, 1.5, 3)
V1 out 0 dc {vo}
R1 out 0 1k
.control
  optimize -dparam xc 4.0 3 7 -center -lhs -samples 120 -analysis op -spec v(out)
  ↪ -max 6 -min 4 -seed 3 -maxiter 40
  print dcenter_yield dcenter_cpk
.endc
```

optimize: design centering -- 1 design param, 1 spec, 120 Latin-Hypercube MC samples/eval, analysis case Cpk

optimize: centered -- worst-case Cpk = 0.6705, yield = 95.83% (120 MC samples), after 39 evaluations
xc = 4.99907

For an output $\sim N(xc, 0.5)$ with spec $[4, 6]$ the optimum is the midpoint 5 and the analytic Cpk $1/(3 \cdot 0.5) = 0.667$; the off-centre start at 4.0 had ~50 % yield.

Worst-case distance (E-305): `wcd` reports the shortest distance, in *standardised* parameter space, from the nominal point to a spec boundary — the sigma level at which the design first fails, and the most probable failure point that gets there (§7.6 shows a run). It answers "how much margin do I have, and in which direction is it thinnest", which a yield percentage cannot: MC estimates the tail by sampling it, and the tail is exactly where samples are scarce. Its dimensions are the netlist's Gaussian `.params` and, under `.option osdimc`, every Gaussian (`* std *`) parameter of every card and instance; `-is <N>` refines the first-order estimate by mean-shift importance sampling centred on the MPFP.

Global optimizers (E-194–E-196) — particle-swarm (pso), differential evolution (de), simulated annealing (sa) — join Nelder–Mead and Levenberg–Marquardt for a multi-modal cost surface, and multi-objective `nsga2` (E-216) returns a Pareto front where the objectives genuinely trade off.

11. Practical guidance

- Reproducibility. Every run is seeded (`setseed`, or the command's `seed/ -seed`); the same seed reproduces the sample sequence bit-for-bit, so results are portable and regressions are detectable.

Without **-seed** the seed is 1, every time — so "run it again" returns the same netlist draws, which is what a paired comparison across design changes wants and what a replication does not. The banner states the seed, and `montecarlo_seed` / `highsigma_seed` / `wcd_seed` publish it for scripts. On the model side `.option mcseed` plays the same role, and inside a loop command `-seed` keys both channels at once.

- Which channel? Netlist `.param` draws when the variation is a property of *this deck* (a tolerance on one resistor, a correlated pair through `mccorr`); model attributes when it is a property of *the device*, shipped with the model and drawn without any deck plumbing; `$rdist_*` when the randomness belongs inside the model's own equations.
- Which sampler? Use plain MC to sanity-check; **`mcsample lhs`** (or `montecarlo -lhs`) for an accurate mean/stddev/yield at a fixed budget; **`highsigma`** when the interesting probability is out in the 4–6 σ tail; **`wcd`** for the margin and its direction.
- Choosing λ for `highsigma`: ~ 2 for 3 σ , ~ 2.5 for 4 σ , ~ 3 for 5–6 σ . Watch the reported relative error — if it is large, raise N or λ — and the effective sample size: under `.option osdimc`, scope the inflation with `-inflate` to the parameters the metric turns on, or the weight collapses.
- Correlations matter. For matched devices, a plausible correlation is worth far more than more samples: as the yield figure shows, ρ can move the yield by tens of percent. On the model side the split is free — model parameter = process, `(* type="instance" *)` = mismatch.
- Bounded parameters. A gauss on a from `(0:inf)` parameter fails trials at the bound; declare `dist="lognormal"` or a `trunc=` instead of sizing the sigma down.
- Keep the record. `.option savemc` costs nothing and makes every sample reconstructible; `writemc` puts the results beside the causes.
- Cost. `montecarlo`'s fast path re-evaluates only the random bindings, and a model-side trial is a parameter write, so a yield/rare-event run is thousands of cheap re-solves — far cheaper than the 10^7 – 10^9 plain-MC runs a 5- σ estimate would otherwise need.

Reproducing the figures

`make_statistics_figs.py` (in this folder) regenerates every plot from real ngspice runs — it drives the committed binary through the `mcsample`, `highsigma`, `mccorr` and `montecarlo` commands, compiles the two Verilog-A fixtures with the committed `openvaf-r` for the `.option osdimc` figures, and plots the results with `matplotlib`. See the per-feature example suites for the verifiers: [lhs_examples](#), [highsigma_examples](#), [yield_examples](#), [montecarlo_examples](#), [mcpolicy_examples](#), [osdimc_examples](#), [osdidist_examples](#), [mctrack_examples](#), [mcyield_examples](#), [savemc_examples](#) and [writemc_examples](#) — the last two each carry a worked example (`mc_example5`, `mc_example6`) with the csv it wrote and the plots made from it.